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SKETCHXAI: EXPLAINABLE AI FOR HUMAN SKETCHES - A HUMAN-CENTERED FRAMEWORK FOR STROKE-LEVEL TRANSPARENCY

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Abstract— SketchXAI is an explainable AI framework that enhances transparency in sketch classification by introducing Stroke-Level Importance (SLI), which identifies the contribution of individual strokes to model predictions. The system combines a CNN trained on the QuickDraw dataset (87 classes, 5,000 samples per class) with LIME-based visual explanations. Evaluation with 63 participants showed 80% explanation accuracy, 77.9% transparency ratings, and 80% user engagement, demonstrating improved trust and interpretability in sketch-based human-AI interaction.

Keywords— Explainable AI, Human Sketch, Stroke-Level Importance, LIME, Convolutional Neural Network.

I. Introduction

As AI systems become more powerful, their decision-making processes become less transparent, creating challenges for user trust and understanding in human-computer interaction (HCI) [1]. Sketch recognition models achieve high accuracy, but lack interpretability; existing XAI methods inadequately explain stroke-based sketch classifications [2]. SketchXAI introduces Stroke-Level Importance (SLI), a method that explains stroke contributions to sketch classifications. Evaluated with 63 participants, it improves transparency, accuracy, and user engagement, while showing potential for psychological assessment applications.

II. Methodology

SketchXAI models sketches as ordered stroke sequences and trains a CNN on the QuickDraw dataset (87 classes, 5,000 samples per class). The network, built with TensorFlow/Keras, achieved 77.87% validation accuracy. To improve interpretability, the framework introduces Stroke-Level Importance (SLI), which measures each stroke's contribution through recovery and transfer mechanisms. Stroke importance is determined by changes in prediction confidence when strokes are modified or removed. Additionally, LIME generates pixel-level heatmaps, providing complementary visual explanations that highlight influential sketch regions alongside stroke-level insights. Figure 1 shows the complete system and methodological pipeline of the SketchXAI framework.

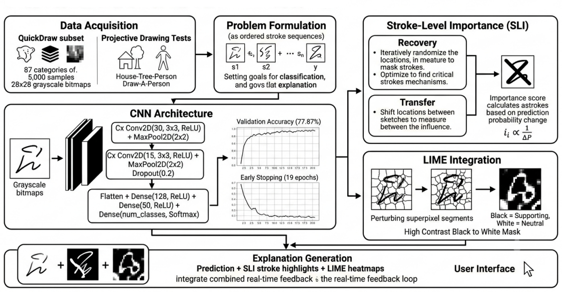

Figure 1. Full System and Methodological Pipeline of the SketchXAI Framework.

III. Results and Discussion

The CNN model was trained on 87-class QuickDraw data. Initial experiments with all 345 classes yielded a test accuracy of 0.67 with evident overfitting (train accuracy 0.66, validation accuracy 0.65). Reducing the dataset to 87 classes with 5,000 samples per class and adding a Dropout(0.2) layer significantly improved generalization. After 19 epochs with early stopping, the model converged to a final validation accuracy of 77.87%, with training and validation loss curves plateauing closely together.

Table 1. Model Performance Comparison

Configuration	Val. Accuracy	Val. Loss
345 classes (baseline)	0.65	1.43
87 classes + Dropout	0.779	0.81

Table 1 presents a comparison between the baseline configuration (345 classes) and the proposed configuration (87 classes with dropout). The proposed model achieves superior performance, increasing validation accuracy from 0.650 to 0.779 while reducing validation loss from 1.43 to 0.81. These results demonstrate the effectiveness of class reduction and dropout regularization in improving model accuracy and generalization.

IV. Conclusion

This paper introduced SketchXAI, an explainable AI framework for sketch classification. By combining Stroke-Level Importance (SLI) and LIME-based visual explanations, the framework achieved 77.87% validation accuracy while improving transparency and interpretability. These results highlight the potential of SketchXAI for trustworthy human-centered AI applications.

References

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